

Quiz (Carolina Reaper)

- Q1.** A shipping company has two types of trucks. Let x be the number of large trucks and y be the number of small trucks. The company has 12 trucks in total and the large trucks are 3 times more expensive than the small trucks. The total cost of the trucks is
- 240,000
- . Which system represents the situation?
- (A) $x + y = 12$, $x = 3y$
(B) $x + y = 12$, $3x + y = 240$
(C) $x + y = 12$, $3x + y = 240,000$
(D) $x - y = 12$, $3x + y = 240,000$
(E) $x + y = 12$, $x = 3y$, $3x + y = 240,000$
- Q2.** Solve the system of equations using substitution: $x - 2y = 3$, $y = 2x - 5$. What is the value of x ?
- (A) $x = 1$
(B) $x = 3$
(C) $x = 7$
(D) $x = 11$
(E) $x = 13$
- Q3.** A traffic flow problem can be represented by the system: $2x + 3y = 12$, $x - 2y = -3$. Which method would be most efficient to solve this system?
- (A) Graphing
(B) Substitution
(C) Elimination
(D) Matrices
(E) Guess and check
- Q4.** A delivery company has 15 drivers and 20 trucks. Let x be the number of trucks that are not in use and y be the number of drivers that are not assigned to a truck. The company has 5 more trucks than drivers. Which equation represents the situation?
- (A) $x + y = 15$
- (B) $x + y = 20$
(C) $x + y = 35$
(D) $x - y = 5$
(E) $x - y = -5$
- Q5.** The system of equations $x + 2y = 7$, $3x + 6y = 21$ has which type of solution?
- (A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions
- Q6.** A logistics company has a system of equations representing the cost of shipping: $2x + 3y = 120$, $x - 2y = -30$. What is the value of y ?
- (A) $y = 10$
(B) $y = 15$
(C) $y = 20$
(D) $y = 25$
(E) $y = 30$
- Q7.** The system $x + y = 4$, $3x - 3y = 12$ has which type of solution?
- (A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) No solution, the equations are parallel
- Q8.** Solve the system using elimination: $x + 4y = 11$, $3x - 4y = 5$. What is the value of x ?
- (A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Open Ended Questions (FRQ)

- Q9.** A shipping company has two types of containers: large and small. Let x be the number of large containers and y be the number of small containers. The company has 25 containers in total and the large containers are 2 times more expensive than the small containers. The total cost of the containers is
- 10,000
- . Write a system of equations to represent the situation and solve for x and y . Show all work and explain your answer.
- Q10.** A traffic flow problem can be represented by the system: $x + 2y = 15$, $3x - y = 10$. Solve the system using substitution or elimination and explain which method you chose and why. Show all work and write a brief conclusion about the solution.

Chef's Tips

- Tip 1: Ensure students show all work and explain their answers
- Tip 2: Encourage students to use substitution or elimination to solve systems of equations
- Tip 3: Remind students to check their answers by plugging them back into the original equations